CS259-S: V-Cache and the Effects of AMD's X3D Implementation

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***Abstract*—CPU architecture shifted to a multi-core implementation once the heat and power wall became a ballooning problem during the early to mid-2000s. Instead of focusing on raw clock frequency, CPU architects directed their efforts to multi-core architecture. CPUs with multiples cores have been implemented to take advantage of key concepts in parallel processing to boost productivity in computers. However, at the same time, CPUs significantly outpaced the speedup of main memory creating a memory-bound problem. AMD, in addition to creating CPUs with multiple cores, made attempts to address the memory-bound problem by pioneering a new cache architecture implementation called V-Cache. V-Cache, which has been show to expand the L3 cache of a CPU by up to 4 times compared to previous iterations, allows for productivity to be boosted because a CPU can read more data at once. This advancement has placed AMD chips at the top of the consumer market as a result of the gains from their novel architecture design.**

***Index Terms*—Parallel Processing, CPU Architecture, V-Cache, L3 Cache**

I. Introduction

The start of the twenty-first century witnessed technological leaps of computing power with the implementation of multiple cores on a singular die within a Central Processing Unit (CPU). As the heat and power wall became an ever-increasing problem to deal with, the days of pursuing raw clock frequency to achieve more cycles per second (Hz) came to an end. The industry, as a response to the heat limitations, turned to implementing a multi-core CPU capable of outputting the same amount of instructions per cycle (IPC) at lower frequencies. A general equation regarding the proportional relationship between power, voltage, and frequency within CPUs can be seen below:

$$P \propto V^2 \times F,$$
$$V \propto F,$$
$$\implies P \propto F^3.$$

Thus, a CPU with a single-core whose frequency is $2f$ will require $8f^3$ power, in comparison to a CPU that is dual-core (two cores), each at frequency f, will only need $2f^3$ power, a quarter of the power necessary to achieve the same performance [1].

Even as the industry has scaled their CPUs, especially amongst desktop consumers, to include 4-core, 8-core, 16-core and even 24-core CPUs, however, limitations from the software side still exist. This phenomenon is known as Amdahl's law, shown in Fig. 1.

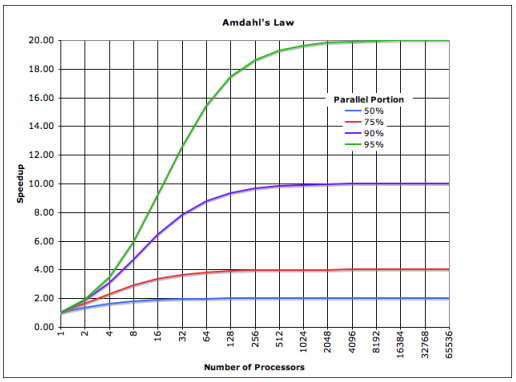


Fig. 1: Speedup factor depending on the percentage of parallelized code.

In essence, regardless of the number of processors that are on a CPU, if the software that is being executed isn't parallelized to take advantage of the multiple cores, the speedup improvements cease to exist. With the intrinsic nature of many applications needing to be sequential, only so much of the source code can be parallelized and so diminishing returns are experienced from more cores. This second bottleneck, Amdahl's Law, in addition to the heat and power wall, caused CPU architects to instead navigate through a different avenue which looks at the memory-bounded problem that exists with all CPUs from today [2].

A memory-bound is a bound measured by the number of cycles the execution pipeline is stalled awaiting data loads, in other words "the time something takes to complete execution is decided primarily by the amount of free memory required to hold working data" [3][4][5]. On the contrary, a CPU-bound is one where a program will execute faster if the CPU is faster. These two overlap with one another but for the sake of this paper, we will only focus on the memory-bound aspect that is present within CPUs.

Building upon memory-bound limitations, there can exist different types of limitations such as one that relies

on memory bandwidth, i.e. how fast data can transfer that is primarily seen in video rendering, graphics design, and other production related tasks. Another limitation is latency, i.e. how fast data can be accessed, which can be found in tasks that are sequential in nature and can't take the full advantage of parallel processing, such as gaming.

There are different types of memory that exist within a standard computer: cache memory, which usually exists as Static Random Access Memory (SRAM), main memory, which usually exists as Dynamic Random Access Memory (DRAM), and virtual memory which is stored on hard drives.

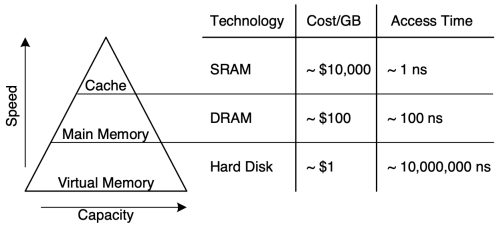


Fig. 2: A hierarchy showing the different types of memory, their cost, and access time [6].

Cache memory is a form of memory that exists on the CPU itself. Most desktop multi-core CPUs have it divided into 3 sections, Level 1 (L1), Level 2 (L2), and Level 3 (L3). Each core has its own L1 and L2 cache. L3 is shared amongst all cores. As the level of memory increases, the distance from the cores increase with the benefit of having larger memory. However, this comes at the expense of memory bandwidth and latency [6].

When the CPU looks for data and finds it in the cache, it is returned quickly and this is referred to as a cache hit. However, when the CPU doesn't find the data in the cache, it must seek the data in the main memory known as DRAM, sometimes referred to simply as RAM. This is called a cache miss [6].

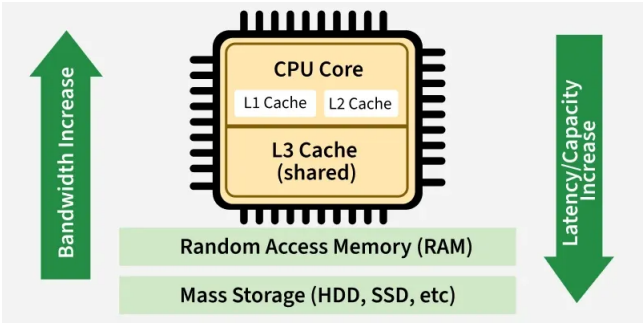


Fig. 3: The relationship between the different levels of cache [7].

DRAM is a type of memory that exists off the CPU and stores bits of data on memory cells. Because it exists off the CPU, it is 1-2 orders of magnitude slower in comparison to cache but has much larger storage [6]. For this reason, cache misses are crucial to a CPU's IPC because cache misses will contribute to CPU idle time, slowing down the system overall.

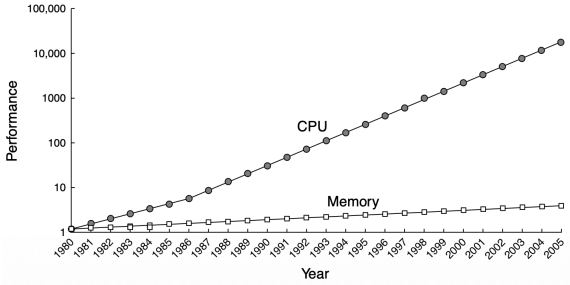


Fig. 4: The speedup of CPU processors compared to main memory [7].

In the 1970s and 1980s, DRAM was able to keep up with up with processors but has since experienced throttles to their speedups as shown in Figure 4. Because of this, increasing the amount of DRAM that exists in a computer system very quickly hits a plateau regarding performance unless the computing task requires large amounts of data.

For reference, DRAM in most computers operates at similar frequencies to a processor, ranging from 1333 MHz up to 3000 MHz for Double Date Rate (DDR) RAM. The bandwidth that DDR RAM is capable of in regards to data transfer rate is upwards of 68 GB/s [8]. Cache speeds, however, are capable of data transfer rates up to 1 TB/s depending on which level of cache is being discussed [9]. Thus, there are a lot of improvements to be made to bridge the gap between main memory speeds and cache memory speeds. AMD has been implementing one method, which they call Vertical Cache (V-Cache), to help their CPUs perform better by targeting the memory bounded problem within them.

II. What is V-Cache?

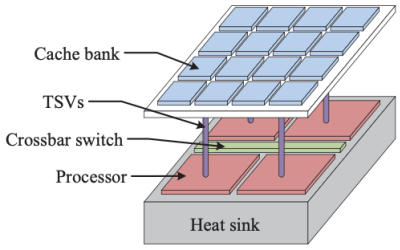


Fig. 5: A depiction of how cache is layered using TSVs [10].

V-Cache, more commonly known as stacked-cache or stacked-memory, is a form of cache that diverts from the traditional method of 2-D implementations and instead takes a 3-D approach by stacking memory modules directly on top of one another. By doing so, we reduce the need to enlarge the die on a CPU, which would introduce latency, and instead maintain the same size and horizontal distance

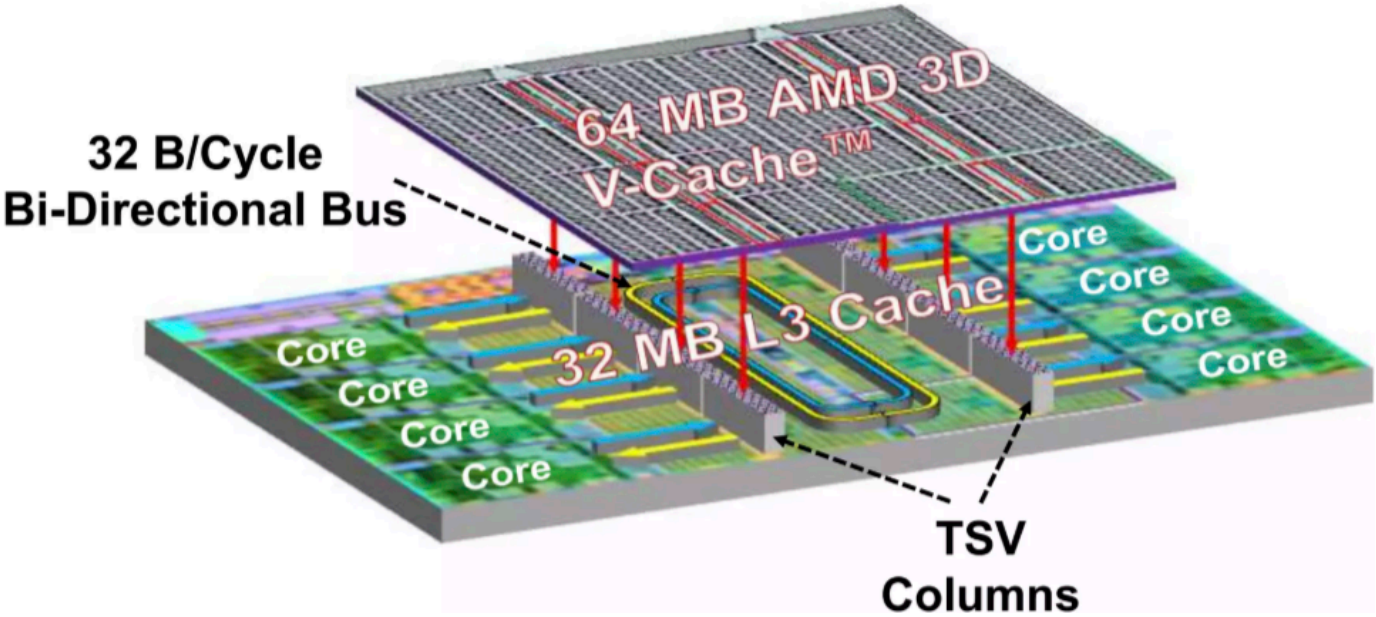


Fig. 6: AMD's implementation of vertical cache [11].

to the cores [10]. This is possible through the creation of Through-Silicon Via (TSV). TSVs are vertical pipes that serve to interconnect dies that are stacked on each other, which allows for performance improvements because of its extremely short wiring length [12].

In 2020 with the release of the Ryzen 7 5800X3D, AMD managed to stack an additional 64 MB of L3 cache onto their processor's die, resulting in a total size of 96 MB of L3 cache [11]. This was the first desktop CPU released with V-Cache implemented. By having more L3 cache, the IPC of a CPU is directly uplifted at a linear scale according to Wuu et. al [11].

The process of attaching an additional layer of L3 cache to a processor's die regarding AMD CPUs is known as hybrid bonding. Hybrid bonding, compared to other methods of vertical cache implementations, offers upwards of 15 times more interconnect density and upwards of 3 times interconnect energy efficiency. The implementation also allows for better thermal conductance [11].

As for benchmarks, in comparison to their non-vertical cache variants, in tasks like gaming, AMD's 5800X3D, which has 8 cores on one die with 96 MB of L3 Cache, performed roughly 15% better than their 5900X, which has 12 cores spread across 2 die and 32 MB of L3 Cache on each die, totaling 64 MB of L3 cache [11][13][14]. Although there are more total cores on the 5900X with two-thirds of the L3 cache, the 5800X3D performs better because games receive diminishing returns at higher core counts and because the 8 cores on a single die have access to 96 MB of cache as opposed to the 5900X's 6 cores having access to 32 MB of cache within each die [15].

III. Limitations of V-Cache

Although V-Cache has shown to be beneficial in terms of IPC and gaming performance, the implementation of the cache and how it is structured is important. A higher total L3 cache does not necessarily outright mean better performance. When looking at the latest AMD Ryzen CPUs, the 9800X3D still maintains the same amount of L3 cache as the 2 generation-old 5800X3D at 96 MB while the 9950X3D is marketed at 128 MB of L3 cache in addition to having double the core count, 16 [16]. However, despite having an additional 32 MB of L3 cache and 8 more cores, the two CPUs perform roughly the same in gaming scenarios [17].

Stephen Burke and his team at Gamer's Nexus ran the 9950X3D through a series of rigorous benchmark testing and compared it to a multitude of CPUs, including the 9800X3D. What they found was that, despite the additional hardware, the two CPUs performed the same across a multitude of different games and in some instances, the 9800X3D would perform slightly better [17]. The difference between the two is highlighted in production tests such as blender rendering, chromium code compile, 7-zip compression, etc. where the tasks being done benefit from having a higher core count and the 9950X3D shines against the 9800X3D[17].

Referring back to the two Ryzen CPUs, the reason why the 9800X3D and 9950X3D perform the same is similar to what was discussed earlier. The 9800X3D has 1 die with 8 cores and 96 MB of L3 cache while the 9950X3D has 2 die with 8 cores each, but only one of the die has 96 MB while the other has 32 MB of L3 cache, totaling to the marketed 128 MB of L3 cache [16]. If AMD wishes to continue getting the best performance uplift in all areas

TABLE I: Intel and AMD CPUs and cache sizes

Name	L1 cache (per core)	L2 cache (per core)	L3 (shared)	Release Year
i7 6700K	64 KB	256 KB	8 MB	2015
i9 7900X	64 KB	1 MB	14 MB	2017
i9 9900K	64 KB	256 KB	16 MB	2018
i9 10900K	64 KB	256 KB	20 MB	2020
i9 11900K	80 KB	512 KB	16 MB	2021
i9 12900K	80 KB	1.25 MB	30 MB	2021
i9 13900K	80 KB	2 MB	36 MB	2022
i9 14900K	80 KB	2 MB	36 MB	2023
Ultra 9 285K	192 KB	3 MB	36 MB	2024
AMD FX 9590	384 KB (total)	8 MB (total)	8 MB	2013
Ryzen 7 1800X	96 KB	512 KB	16 MB	2017
Ryzen 9 3950X	64 KB	512 KB	64 MB	2019
Ryzen 9 5950X	64 KB	512 KB	64 MB	2020
Ryzen 9 7950X3D	64 KB	1 MB	128 MB	2023
Ryzen 9 9950X3D	80 KB	1 MB	128 MB	2025

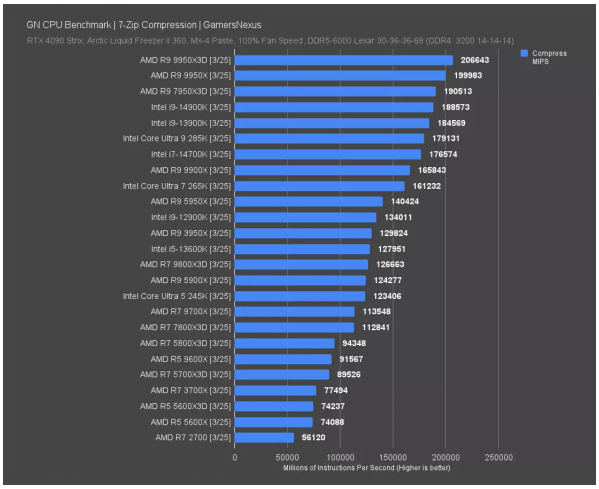


Fig. 7: 7-Zip Benchmark of multiple CPUs ran by Gamer's Nexus [17].

and not just production tasks, the cache size on a singular die must increase as opposed to the total sum of L3 cache across multiple die.

Another limitation that comes with increasing the cache size on a CPU is the power requirements that come with it. Barret, Penhoat, Picard, and Chillet did an empirical analysis related to the estimation of energy costs of data transfers in cache and what they found was that roughly 30% percent of a CPU's power on a CPU like Intel's Xeon server CPU is used by the large L3 cache size [18]. This is due to data movements across memory hierarchies, such as L1 to L3, and even DRAM. In addition to this, prefetching mechanisms, a method to fetch data before it is needed, also further increases power consumption and leads to a higher Thermal Design Power (TDP) [18].

Based on the evidence presented, we can conclude that although AMD is solving a portion of the memory wall problem by decreasing the number of cache misses that occur by enlarging the L3 cache size on their CPUs, the misses that do occur still tax the CPU's IPC. Until computer architects bridge the gap between processor speeds and main memory speeds, shown in Fig. 4, CPUs will always be bottlenecked by main memory speeds.

IV. AMD Post X3D Reveal & Motivations Behind V-Cache Research

Between the years of 2007 to 2017, Intel dominated the market in CPU performance amongst desktop users [19]. With every new generation released, Intel maintained a healthy lead against AMD until the introduction of the Ryzen processors. With the Ryzen series, AMD reduced the lead with their 1000 and 3000 series CPUs. Eventually they overtook Intel in single-threaded CPU performance with the release of 5950X in 2020 when compared to Intel's latest CPU, a benchmark that Intel held the lead in for over a decade. This officially meant the title of best performing CPUs now belonged to AMD [19]. With AMD CPUs finally surpassing the performance of their Intel counterparts, the introduction of their V-Cache architecture with the X3D chips only widened the gap.



Fig. 8: A market cap comparison of the last 10 years between Intel and AMD.

Comparing each generation of CPUs from Intel and AMD in table I, we can see the gap between L3 caches widening between 2017 and 2023. In roughly the same time period, AMD overtook Intel in market capitalization, indicating that more people, gamers and non-gamers alike,

were growing confident and investing in AMD because of the leaps made in their computing power.

This should serve as an indicator on what Intel can focus on if they want to remain competitve in the desktop CPU market. While there hasn't been any confirmation, rumors indicate that, as of Q1 2026, Intel is working on a big Last Level Cache (bLLC) for their CPUs as a response to AMD's V-Cache [20].

V. Conclusion

Reflecting upon the memory hierarchy, how different types of memory work together, where the bottlenecks form, and the ever-increasing memory wall problem that we are facing as processors become significantly faster than main memory, we've seen how AMD has tried to address this problem with the implementation of V-Cache. V-Cache, a 3D implementation of cache architecture, greatly expands the size of the L3 cache without introducing additional latency via distance by strategically stacking the cache directly on top of the die with the usage of TSVs. This method has been proven to increase a CPU's IPC by reducing the number of cache misses that occur, improving overall performance.

However, power consumption from increasing the size of L3 cache can pose a potential barrier in the future. The implementation of the 3D cache on certain die only has also been shown to limit performance where multiple die (cores) are not needed.

Overall, the implementation has placed AMD as the industry leader amongst desktop consumers and enthusiasts in comparison to Intel and their lack of a similar implementation. Intel, as a response, has been rumored to attempt to increase L3 cache in a similar capacity with bLLC.

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